Fourier Series

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Fourier Series over an interval $(\alpha, \alpha + 2l)$ is given by:

$$f(x) \sim \frac{1}{2l}\int_{\alpha}^{\alpha+2l} f(x)dx + \sum_{n=1}^{\infty} \cos(\frac{n\pi x}{l})\frac{1}{l}\int_{\alpha}^{\alpha+2l} f(x)\cos(\frac{n\pi x}{l})\,dx + \sum_{n=1}^{\infty} \sin(\frac{n\pi x}{l})\frac{1}{l}\int_{\alpha}^{\alpha+2l} f(x)\sin(\frac{n\pi x}{l})\,dx$$

Fourier Half Range Cosine Series and Half Range Sine Series are respectively given by:

$$f(x) \sim \frac{1}{l}\int_{0}^{l} f(x)dx + \sum_{n=1}^{\infty} \cos(\frac{n\pi x}{l})\frac{2}{l}\int_{0}^{l} f(x)\cos(\frac{n\pi x}{l})\,dx \;;\; f(x) \sim \sum_{n=1}^{\infty} \sin(\frac{n\pi x}{l})\frac{2}{l}\int_{0}^{l} f(x)\sin(\frac{n\pi x}{l})\,dx$$

Solution of wave Equation:

$$\frac{\delta^2 y}{\delta t^2} = c^2\frac{\delta^2 y}{\delta x^2} \Rightarrow y(x,t) = [C_1\cos(px) + C_2\sin(px)][C_3\cos(cpt) + C_4\sin(cpt)]$$

<u>Boundary</u>: $y(0,t) = 0, y(l,t) = 0$ <u>Initial</u>: $y(x,0) = f(x), \frac{\delta y}{\delta t}(t=0) = g(x), 0 \le x \le l$

Solution of Heat Equation:

$$\frac{\delta y}{\delta t} = c^2\frac{\delta^2 y}{\delta x^2} \Rightarrow y(x,t) = [C_1\cos(px) + C_2\sin(px)]e^{-c^2p^2t}$$

<u>Boundary</u>: $y(0,t) = T_1, y(l,t) = T_2$ <u>Initial</u>: $y(x,0) = f(x), 0 \le x \le l$

Convergence of Fourier Series:
Let we have a Fourier Series $FS(x) \sim f(x), \alpha \le x \le \beta$

$$FS(x) \to \begin{cases} \dfrac{f(k^+) + f(k^-)}{2}, & at\ x = k \in [\alpha, \beta] \\ \dfrac{f(\alpha^+) + f(\beta^-)}{2}, & at\ x = \alpha, \beta \end{cases}$$